H.T.No. | | | | | | | | | | |

Code No: CT4516 **SRGEC-R23**

III B.Tech I Semester Regular Examinations, November 2025

ARTIFICIAL INTELLIGENCE

(Computer Science and Engineering & Information Technology)

Time: 3 Hours **Max. Marks: 70**

Note: All questions from **PART-A** are to be answered at one place.
Answer one question from each unit of **PART-B.** All questions carry equal marks.

PART-A

10 × 2 = 20M

1. a) Draw the diagram for how agent can interact with environments through sensors and actuators.
   b) List out the properties of the task environments.
   c) What problems will be faced in game playing algorithms?
   d) What is meant by Game Playing-Adversial search
   e) What are the disadvantages of prepositional logic? How they can be reduced by using predicate logic?
   f) Define semantic networks in the context of AI. What are the key elements that make up a semantic net?
   g) What is reinforcement learning?
   h) List the main components of a decision tree in inductive learning.
   i) Write down the applications of expert system.
   j) what is expert systems shells.

PART-B

5 × 10 = 50M

UNIT-I

2. a) What is meant by Rationality and Rational Agent? Explain how agents can interact with environment using sensors and actuators? (5M)
   b) List and explain the components that are involved in problem formulation. (5M)

(OR)

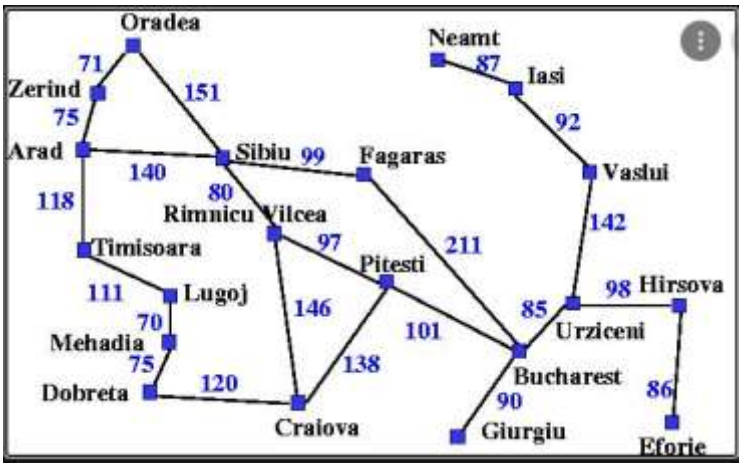
3. a) Show that the 8 Queens problem can be classified under the area of Artificial Intelligence. Describe the incremental problem formulation of this problem. (5M)
   b) Write a brief description about AI problems. (5M)

UNIT-II

4. a) In what kind of problems the Breadth first search be better than a depth first search? (4M)
   b) With an example illustrate the use of Alpha – Beta cutoffs in minimax procedure. (6M)

(OR)

5. For the following example which consists of straight line distances for Bucharest, By applying A* algorithm on this instance show the expanded stages of each node by taking initial state as Arad and label each node with f= g+h. (10M)



City	$h_{(SLD)}$	City	$h_{(SLD)}$
Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

UNIT-III

6. a) How the ISA and INSTANCE relationships can be represented in predicate logic? Discuss with an example. (5M)

   b) What are the approaches to knowledge representation? Discuss. (5M)

(OR)

7. Develop a frame-based system for representing knowledge about a university system (e.g., courses, students, professors). Define the frames and slots for each entity and demonstrate how this system can be used to infer information. (10M)

UNIT-IV

8. a) Write about Forward chaining algorithm with example. (5M)

   b) Describe the functioning of inducing decision trees. (5M)

(OR)

9. a) How can an agent learn from success and failure & from reward and punishment? Discuss. (5M)

   b) Explain the inference rules for quantifiers. (5M)

UNIT-V

10. Compare and contrast Expert system and traditional system. (10M)

(OR)

11. Analyze the knowledge representation techniques used in MYCIN and compare them to modern expert systems. (10M)

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